

2024/2025 Second Semester

SC2005 Operating Systems

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Learning Outcomes

- Identify the functions and services provided by an operating system
- Understand the basic concepts and mechanisms of operating system design and implementations
- Explain the interactions between different components of an operating system

Course Components

- **Lectures:** 13 weeks, 2 hours/week
- **Tutorials:** 12 tutorials
- **Labs:** 4 lab sessions (experiments), 2 hours/session
- **Assessment:** Coursework (**100%**)
 - Lecture quizzes (**12%**) – 8 multiple-choice quizzes
 - Lab implementations (**18%**)
 - Lab quizzes (**25%**) – multiple-choice quizzes
 - Concept quiz (**15%**) – multiple-choice quiz
 - Algorithm quiz (**30%**) – short answers
 - BONUS: Tutorial attendance can give **5%** extra

Schedule (Full-Time Students)

SC2005 Operating Systems

Schedule for Semester 2, Academic Year 2024/2025

Full-time Program

Week	Lecture Topic	Tutorial	Lab	Assessments Due
1	13/01 - 17/01	Overview of Operating Systems (OS)		
2	20/01 - 24/01	Processes & Thread	1 (OS Overview)	
3	27/01 - 31/01	Process Scheduling	2 (Processes & Threads)	Lecture Quiz 1
4	03/02 - 07/02	Process Synchronization	3 (Process Scheduling)	Lecture Quiz 2
5	10/02 - 14/02	Process Synchronization	4 (Process Synchronization 1)	1 (Nachos Thread)
6	17/02 - 21/02	Deadlocks	5 (Process Synchronization 2)	1 (Nachos Thread)
7	24/02 - 28/02	Real-time OS & Virtualization	6 (Deadlocks)	2 (CPU Scheduling)
	03/03 - 07/03	Recess Week		Lecture Quiz 3
8	10/03 - 14/03	Memory Organization	7 (Real-Time OS & Virtualization)	2 (CPU Scheduling)
9	17/03 - 21/03	Memory Organization / Virtual Memory	8 (Memory Organization)	3 (CPU Scheduling)
10	24/03 - 28/03	Virtual Memory	9 (Virtual Memory 1)	1 (Nachos Thread)
11	31/03 - 04/04	File Systems	10 (Virtual Memory 2)	Lecture Quiz 2
12	07/04 - 11/04	Input/Output (I/O) Management & Disk Scheduling	11 (File Systems)	Lecture Quiz 3
13	14/04 - 18/04	Protection & Security (e-learning)	12 (I/O & Disk Scheduling)	Lecture Quiz 4
14	21/04 - 25/04	Quizzes		Lecture Quiz 5 & Lab Quiz 1

Textbook: Operating System Concepts, by Silberschatz, Galvin and Gagne, 10th Edition, Wiley 2018

Public Holidays: 29 Jan 2025 (Wednesday), 30 Jan 2025 (Thursday), 31 Mar 2025 (Monday), 18 Apr 2025 (Friday)

- Tutorial starts from Week 2
- Lab starts from Week 4
- For specific deadlines, please refer to Assessments folder in NTULearn

Schedule (Part-Time Students)

Part-time Program

Week		On line Lecture (2 Hours)	Tutorial (6:30PM-7:30PM) SWLab1	Lecture Inq (7:30PM-8:30PM) SWLab1	Lab (7:30PM-9:30PM) SWLab1	Assessments Due
1	13/01 - 17/01	Overview of Operating Systems (OS)				
2	20/01 - 24/01	Processes & Threads	1 (OS Overview)	1 (OS Overview)		
3	27/01 - 31/01	Process Scheduling				Lecture Quiz 1
4	03/02 - 07/02	Process Synchronization	2 (Processes & Threads), 3 (CPU Scheduling)	2 (Process Concept and Scheduling)		Lecture Quiz 2
5	10/02 - 14/02	Process Synchronization	4 (Process Synchronization 1)		1 (Nachos Thread)	
6	17/02 - 21/02	Deadlocks	5 (Process Synchronization 2)	3 (Process Synchronization)		Lecture Quiz 3
7	24/02 - 28/02	Real-time OS & Virtualization	6 (Deadlocks)		2 (CPU Scheduling)	
	03/03 - 07/03	Recess Week				Lecture Quiz 4 & Lab Quiz 1
8	10/03 - 14/03	Memory Organization	7 (Real-Time OS & Virtualization)	4 (Deadlock & Real-time OS)		
9	17/03 - 21/03	Memory Organization / Virtual Memory	8 (Memory Organization)		3 (Process Synchronization)	
10	24/03 - 28/03	Virtual Memory	9 (Virtual Memory 1)	5 (Memory)		Lecture Quiz 5
11	31/03 - 04/04	File Systems	10 (Virtual Memory 2)		4 (Virtual Memory)	Lecture Quiz 6
12	07/04 - 11/04	Input/Output (I/O) Management & Disk Scheduling	11 (File Systems)	6 (File Systems & I/O)		Lecture Quiz 7
13	14/04 - 18/04	Protection and Security (e-learning)	12 (I/O & Disk Scheduling)			Lecture Quiz 8 & Lab Quiz 2
14	21/04 - 25/04	Quizzes				Concept Quiz & Algorithm Quiz

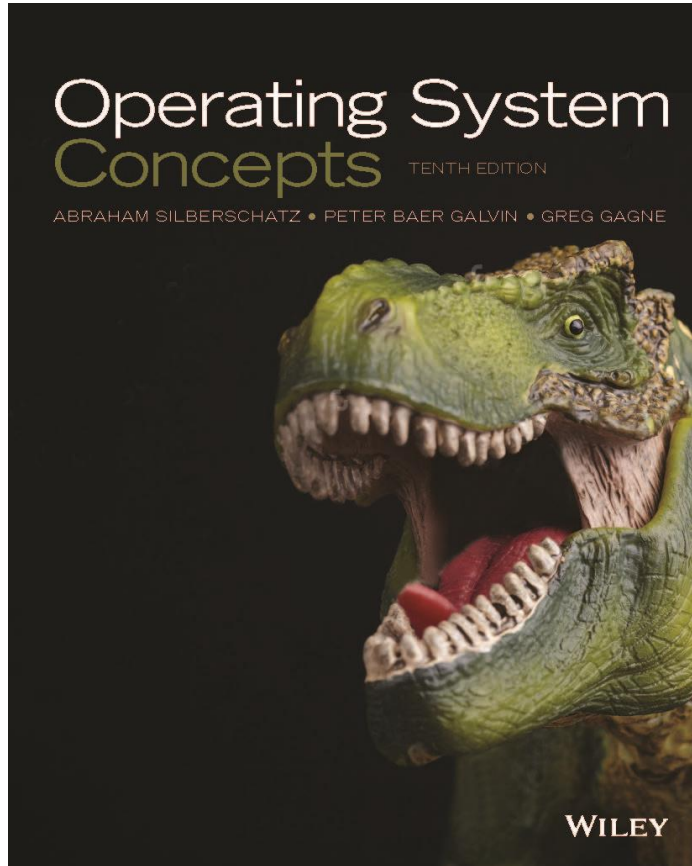
Textbook: Operating System Concepts, by Silberschatz, Galvin and Gagne, 10th Edition, Wiley 2018

Public Holidays: 29 Jan 2025 (Wednesday), 30 Jan 2025 (Thursday), 31 Mar 2025 (Monday), 18 Apr 2025 (Friday)

- Tutorial starts from Week 2
- Lab starts from Week 5
- For specific deadlines, please refer to Assessments folder in NTULearn

Lectures

- Textbook – The *Dinosaur* Book



Textbook: Operating System
Concepts,
by Silberschatz, Galvin and Gagne
10th Edition, Wiley 2018

Textbook vs. Slides

Tutorials

- Start in Week 2
- Attendance is important
- All tutorial questions are available in NTULearn
- Necessary to attempt questions before tutorial
- Some questions are from the previous exams
- Feedback channel is always open
- Outline solutions will be released at the end of Week 13

Labs

- Starts from Week 4
 - Using **Software Lab 1** (N4-01a-02), **Software Lab 2** (N4-01c-06) and **Software Lab 3** (N4-B1c-14); **remote access available**
 - **Attendance is compulsory (makeup only for valid reasons with official supporting document such as MC)**
- 4 lab experiments, 1 per lab session
 1. **NachOS** threads – Introduction to multi-threading
 2. CPU scheduling – Implementation of Round-Robin scheduling
 3. Process synchronization – Importance and techniques
 4. Virtual memory – Implementation of TLB and page replacement
- Lab manuals, briefing slides and other supplementary documents are available in NTULearn
 - **For each experiment, first review the briefing slides and read the lab manual**

Assessments

- Lecture Quizzes (12%)
 - NTULearn tests with multiple-choice quizzes
 - **3 retries allowed** for the quizzes – highest grade recorded
 - Quizzes are released right after the lecture related to the topics, and are due next week
 - Quiz topics
 - Quiz 1: OS intro., processes/threads
 - Quiz 2: Process scheduling
 - Quiz 3: Process synchronization
 - Quiz 4: Deadlocks, Real-Time OS & Virtualization
 - Quiz 5: Memory Organization
 - Quiz 6: Virtual Memory
 - Quiz 7: File Management
 - Quiz 8: I/O Management

Assessments

- Lab Assessments

- Lab Implementation (18%)

- Correct implementation and output generation for each lab experiment – (3% for lab1, 5% for labs 2 - 4)
 - TA or Lab Supervisor will compile/execute your code and evaluate your implementation and output files
 - **Deadline is 1 week after your corresponding lab session**
 - **Remember to leave your implementation and all generated outputs in your folder, and DO NOT modify the folder structure, folder names or file names in anyway**

- Lab Multiple-Choice Quizzes (25%)

- Quiz 1: On Lab 1 and Lab 2 topics – **Recess Week**
 - Quiz 2: On Lab 3 and Lab 4 topics – **Week 13**

Assessments

- Concept & Algorithm Quizzes

- Concept Quiz (15%) – Week 14

- Covering topics in Weeks 1-12
 - Multiple-choice questions focussing on concepts

- Algorithm Quiz (30%) – Week 14

- Covering topics in Weeks 1-12
 - Short answer questions focussing on concepts, problem solving and algorithms

Makeup Policy

- Lecture Quizzes

- ❑ **No makeup is allowed** (students are given 1 week time to complete these quizzes)

- Lab implementations

- ❑ Attendance is compulsory (makeup lab is feasible with a valid document such as MC)
 - ❑ Lab implementations are due 1 week after your lab session (check lab manuals for details)
 - ❑ **Late submissions will attract a penalty of 25%**

- Lab Quizzes, Concept Quiz, Algorithm Quiz

- ❑ **ONE makeup quiz will be arranged for each**
 - ❑ **To be eligible for makeup, a valid MC must be submitted via email on the day of quiz**

Q&A and Email Policy

- **Check the course site and lecture slides in NTULearn**
 - Most info will be available there
- **Use Discussion Board in NTULearn**
 - Course content Q&A and organization Q&A forums
 - Replying to peers is highly encouraged and **not graded**
 - Policy: Be clear, brief, constructive, **respect others**
 - Anonymous Q&A are ok
 - Check the Discussion Board **before** sending emails
- **Sending an email**
 - Add [SC2005] to the **email subject**
 - We **might ignore** emails if the info is available in NTULearn